

Calculus I Lesson 48

Question 1:

$$f(x) = \sinh(12x)$$

Hint: Let $u = 12x$

a) $u' = \frac{du}{dx} = \underline{\hspace{2cm}}?$

b) $f'(x) = \underline{\hspace{2cm}}?$

Hint: Use Chain Rule.

Question 2:

$$f(x) = \operatorname{sech}(15x^2)$$

Hint: Let $u = 15x^2$

a) $u' = \frac{du}{dx} = \underline{\hspace{2cm}}?$

b) $f'(x) = \underline{\hspace{2cm}}?$

Hint: Use Chain Rule

Question 3:

$$f(x) = \frac{1}{2} \sinh(5x) - \frac{x}{2}$$

$$\text{a) } D_x \left(\frac{1}{2} \sinh(5x) \right) = \underline{\hspace{2cm}} ?$$

$$\text{a) } D_x \left(\frac{x}{2} \right) = D_x \left(\frac{1}{2} \cdot x \right) = \underline{\hspace{2cm}} ?$$

$$\text{c) } f'(x) = \underline{\hspace{2cm}} ?$$

Question 4:

$y = \arctan(\cosh x)$ Find equation of tangent line $(0, \pi/4)$

a) $y' = ?$

b) Slope of tangent line $= y'(0) = ?$

c) Equation of tangent line: $?$

Hint: $y - y_1 = m(x - x_1)$

Hint:

$$\frac{d}{dx} [\arctan u] = \frac{1}{1 + u^2} \cdot u'$$

$$\frac{d}{dx} [\cosh u] = (\sinh u) \cdot u'$$

Question 5:

$y = \sinh(4 - x^2)$ Find equation of tangent line (2, 0)

a) $y' =$? _____

b) Slope of tangent line $= y'(2) =$? _____

c) Equation of tangent line: ? _____

Hint: $y - y_1 = m(x - x_1)$

Hint:

$$\frac{d}{dx}[\sinh u] = (\cosh u) \cdot u'$$

Question 6:

Find $\int \cosh 12x dx$.

Hint: Let $u = 12x$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}$

b) $\int \cosh 12x dx = \underline{\hspace{2cm}}$

Hyperbolic Functions

$$\int \cosh u du = \sinh u + C$$

$$\int \sinh u du = \cosh u + C$$

$$\int \operatorname{sech}^2 u du = \tanh u + C$$

$$\int \operatorname{csch}^2 u du = -\coth u + C$$

$$\int \operatorname{sech} u \tanh u du = -\operatorname{sech} u + C$$

$$\int \csc hu \coth u du = -\csc hu + C$$

Question 7:

Find $\int \sinh(10 - x)dx$.

Hint: Let $u = 10 - x$

a) $\frac{du}{dx} = \underline{\hspace{2cm}} ?$

b) $\int \sinh(10 - x)dx = \underline{\hspace{2cm}} ?$

Question 8:

Find $\int \cosh^2(x-10) \cdot \sinh(x-10) dx$.

Hint: Let $u = \cosh(x-10)$

a) $u' = \frac{du}{dx} = ?$

b) Write the integral $\int \cosh^2(x-10) \cdot \sinh(x-10) dx$ in terms of u : $?$

c) $\int \cosh^2(x-10) \cdot \sinh(x-1) dx = ?$

Hyperbolic Functions

$$\int \cosh u du = \sinh u + C$$

$$\int \sinh u du = \cosh u + C$$

$$\int \operatorname{sech}^2 u du = \tanh u + C$$

$$\int \operatorname{csch}^2 u du = -\coth u + C$$

$$\int \operatorname{sech} u \tanh u du = -\operatorname{sech} u + C$$

$$\int \csc hu \coth u du = -\csc hu + C$$

Question 9:

Find $\int \frac{\cosh x}{\sinh x} dx = \int \frac{1}{\sinh x} \cdot \cosh x \cdot dx$

Hint:

$$\frac{d}{dx}[\sinh u] = (\cosh u) \cdot u'$$

Hint: Let $u = \sinh x$

a) $\frac{du}{dx} = ?$

b) Write the integral $\int \frac{1}{\sinh x} \cdot \cosh x \cdot dx$ in terms of u : $?$

c) $\int \frac{\cosh x}{\sinh x} dx = \int \frac{1}{\sinh x} \cdot \cosh x \cdot dx = ?$

Question 10:

$$\text{Find } \int_0^2 \tanh x dx = \int_0^2 \frac{\sinh x}{\cosh x} dx = \int_0^2 \frac{1}{\cosh x} \sinh x dx$$

Hint: Let $u = \cosh x$

a) $\frac{du}{dx} = ?$ _____

b) $\int \tanh x dx == ?$ _____ (indefinite integral)

c) $\int_0^2 \tanh x dx == ?$ _____ (definite integral)